Reward Recovery in Sequential Decision Models

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1 Setup

Consider a stationary infinite-horizon MDP with finite state space $\mathcal{S}$, finite action space $\mathcal{A}$, discount factor $\beta \in (0,1)$, transition kernel $P : \mathcal{S} \times \mathcal{A} \to \Delta(\mathcal{S})$, and unknown reward $r : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$. Additive T1EV shocks yield the soft Bellman recursion.

$$Q^*(s,a) = r(s,a) + \beta \sum_{s'} P(s' \mid s,a)\, V^*(s'), \tag{1}$$

$$V^*(s) = \log \sum_{a'} \exp Q^*(s,a'), \tag{2}$$

$$\pi^*(a \mid s) = \exp\bigl(Q^*(s,a) - V^*(s)\bigr). \tag{3}$$

The advantage $A^*(s,a) = Q^*(s,a) - V^*(s) = \log \pi^*(a \mid s)$ is identified from CCPs. The reward r is not identified in levels: any $\tilde{r}(s,a) = r(s,a) + \lambda(s)$ generates identical CCPs (Magnac and Thesmar, 2002). The set of all such $\tilde{r}$ is the identified set (Kalouptsidi et al., 2021).

2 Four Approaches

2.1 Reduced-form Q

The reduced-form approach estimates $Q_\omega(s,a)$ by logit MLE.

$$\hat{\omega} = \arg\max_{\omega} \sum_{i,t} \log \frac{\exp Q_\omega(s_{it}, a_{it})}{\sum_{a'} \exp Q_\omega(s_{it}, a')}. \tag{4}$$

At the population optimum, $Q_{\hat{\omega}}(s,a) = Q^*(s,a) + c(s)$ for some action-independent c, so the backed-out "reward" is $A^*(s,a)$, not $r(s,a)$. No Bellman equation is imposed and no transition data are used.

2.2 AIRL (Fu et al., 2018)

The AIRL discriminator decomposes as follows.

$$f_\theta(s,a,s') = g_\theta(s,a) + \beta\, h_\phi(s') - h_\phi(s). \tag{5}$$

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Here g_θ approximates the reward and $\beta h(s') - h(s)$ is potential-based shaping (Ng et al., 1999). At the GAN optimum the discriminator score equals the advantage, $f^* = A^*$ (Fu et al., 2018, Appendix A.4), which implies the following expression for the recovered reward network.

$$g^*(s,a) = r(s,a) - \delta(s) + \beta \sum_{s'} P(s' \mid s,a)\, \delta(s'), \tag{6}$$

where $\delta(s) \equiv V^*(s) - h^*(s)$ is the shaping residual. Unless $\delta \equiv 0$, the recovered g^* is a potential-based perturbation of r that generates the same policy in the original MDP but can produce different behavior under changed dynamics.

Fu et al. address this through a state-only restriction: if g_θ depends only on s (not a) and the dynamics satisfy a decomposability condition, then $g^*(s) = r(s) + \text{const}$, recovering a disentangled reward that transfers across dynamics. This is what enables their sim-to-real experiments. However, the state-only restriction requires $r(s,a) = r(s)$, which fails whenever rewards depend on actions (e.g., paying vs. waiting).

2.2.1 LSW anchor identification (Lee et al., 2026)

Lee et al. (2026) (henceforth LSW) take a different approach. They allow action-dependent rewards and impose two anchors instead. First, the exit payoff is normalized to zero, so $g(s, \text{exit}) = 0$ for all s. Second, the shaping function is pinned at the absorbing state, so $h(s_{\text{abs}}) = 0$, where exit deterministically transitions to s_{abs}.

Lemma 1 (Non-identifiability)**.** *If (g,h) and $(\bar{g},\bar{h})$ produce the same f for all (s,a,s'), then $\bar{h} = h + \Phi$ and $\bar{g}(s,a) = g(s,a) + \Phi(s) - \beta\Phi(s')$ for some $\Phi : \mathcal{S} \to \mathbb{R}$.*

Proof. Rearrange $g + \beta h(s') - h(s) = \bar{g} + \beta\bar{h}(s') - \bar{h}(s)$ and define $\Phi = \bar{h} - h$. □ □

Theorem 2 (Anchor identification)**.** *Under the two anchors, if (g,h) and $(\bar{g},\bar{h})$ realize the same f, then $g = \bar{g}$ and $h = \bar{h}$.*

Proof. By Lemma 1, $\bar{g}(s, \text{exit}) = 0 + \Phi(s) - \beta\Phi(s_{\text{abs}})$. The exit anchor forces $\Phi(s) = \beta\Phi(s_{\text{abs}})$ for all s. The absorbing anchor forces $\Phi(s_{\text{abs}}) = 0$. Hence $\Phi \equiv 0$. □ □

Theorem 3 (Recovery of true reward)**.** *If the true pair (r^*, V^*) satisfies $r^*(s, \text{exit}) = 0$ and $V^*(s_{\text{abs}}) = 0$, then any AIRL pair (g,h) realizing A^* under the same anchors satisfies $g = r^*$ and $h = V^*$.*

Proof. Define $\Phi = h - V^*$. The exit anchor gives $\Phi(s) = \beta\Phi(s_{\text{abs}})$. The absorbing anchor applied to both h and V^* gives $\Phi(s_{\text{abs}}) = 0$. □ □

The Kang et al. (2025) ERM framework uses the same anchor principle but solves a single non-adversarial optimization instead of a GAN.

The LSW model estimates segment-specific reward and policy functions for K latent consumer types. Each user belongs to segment $z \in \{1, \ldots, K\}$, and the objective is to recover the segment-specific reward $r^*(s,a,z)$ and policy $\pi^*(a \mid s,z)$ that rationalize observed trajectories under the entropy-regularized Bellman equation. Estimation proceeds via an EM algorithm: the E-step computes posterior segment membership from trajectory likelihoods, and the M-step updates each segment's reward via AIRL with the anchor constraints applied per segment.

2.3 IQ-Learn (Garg et al., 2021)

The inverse Bellman operator $r_Q(s,a) \equiv Q(s,a) - \beta \sum_{s'} P(s' \mid s,a)\, V_Q(s')$ expresses reward as a function of Q alone. IQ-Learn estimates a single Q-function by solving the following optimization problem.

$$\max_Q \; \mathbb{E}_{(s,a)\sim\rho^*}\big[\phi\big(r_Q(s,a)\big)\big] - (1-\beta)\,\mathbb{E}_{s_0\sim\mu_0}\big[V_Q(s_0)\big], \tag{7}$$

The function $\phi(x) = x - x^2/2$ corresponds to χ^2 divergence. At the optimum, r_Q separates immediate reward from continuation values via the Bellman constraint. IQ-Learn requires the transition kernel $P(s' \mid s,a)$ to compute the inverse Bellman operator and uses it in the objective through $V_Q(s')$. It does not require adversarial training.

2.4 GLADIUS (Kang et al., 2025)

GLADIUS parameterizes both $Q_\theta(s,a)$ and $\hat{E}_\theta[V(s') \mid s,a]$ with neural networks, trained jointly on observed (s,a,s') transitions via the following loss.

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_i \log \pi_\theta(a_i \mid s_i) + \lambda \frac{1}{N}\sum_i \big(\hat{E}_\theta[V(s_i') \mid s_i, a_i] - V_\theta(s_i')\big)^2, \tag{8}$$

where $V_\theta(s') = \log \sum_{a'} \exp Q_\theta(s',a')$. After training, reward is extracted via the same inverse Bellman operator as IQ-Learn. Unlike IQ-Learn, GLADIUS does not require the transition matrix P at estimation time. It learns $\hat{E}[V(s') \mid s,a]$ from empirical next-state observations rather than computing it from known transitions. The MLP architecture shares parameters across states, providing implicit regularization that tabular Q parameterizations lack.

3 Counterfactual Taxonomy

Definition 4 (Type I: state extrapolation)**.** *The transition kernel P and reward r are unchanged. Only the realized state values shift. The counterfactual CCPs require only evaluating the estimated Q at new state inputs, with no Bellman re-solve.*

Definition 5 (Type II: environment change)**.** *The reward r is unchanged, but the transition kernel shifts to $\tilde{P}$ or the action set changes. The counterfactual Q must satisfy a new Bellman equation $\tilde{Q} = r + \beta\tilde{P}\tilde{V}$, which requires the structural reward r separated from the old continuation values.*

Proposition 6 (Type I equivalence)**.** *Under Type I, all four methods yield identical counterfactual CCPs.*

Proof. Each method recovers scores whose within-state action differences match Q^*, which can be written as $Q^*(s,a) + c(s)$ for some action-independent c. For any shifted state $\tilde{s}$, the softmax $\pi(a \mid \tilde{s}) \propto \exp(Q^*(\tilde{s},a) + c(\tilde{s}))$ cancels $c(\tilde{s})$. □ □

Proposition 7 (Type II divergence)**.** *Under Type II, reduced-form and AIRL-without-anchors produce invalid predictions. AIRL with anchors, IQ-Learn, and GLADIUS produce valid predictions when the structural reward is correctly recovered.*

The reduced-form estimate $Q^* + c(s)$ bundles r with $\beta P V^*$, so under $\tilde{P}$ it cannot disentangle r. AIRL without anchors recovers $g^* = r + \text{shaping}$, which is observationally equivalent in the original MDP but produces incorrect behavior under $\tilde{P}$ unless $\delta \equiv 0$. AIRL with anchors recovers $g^* = r^*$ exactly (Theorem 3). IQ-Learn and GLADIUS separate r via the inverse Bellman operator. The LSW anchors provide what Kalouptsidi et al. (2021) call the "additional restrictions" needed to identify counterfactuals under transition changes.

4 Application to LSW (2026)

LSW study sequential content consumption with three actions (buy, wait, exit) using AIRL under the two anchors and deterministic transitions. Their wait-time validation (Section 6.3) is a Type I counterfactual, so every method that recovers the advantage function would pass it identically. Their WFF removal and segment-specific delay policies (Sections 7.1–7.2) are Type II because they change the action-set structure and the MDP that each segment faces. Their content-based paywall (Section 7.3) is also Type II because high-content episodes become paid-only, altering the transition structure episode-by-episode.

5 Simulation Study

We construct a tabular MDP that mimics the LSW book-reading environment. The state encodes episode position, content quality, and wait-time level, giving $15 \times 3 \times 4 = 180$ regular states plus one absorbing state. The three actions are buy, wait, and exit, with deterministic transitions and $\beta = 0.95$. The true reward is $r^*(s, \text{buy}) = u(c) - 2.0$, $r^*(s, \text{wait}) = u(c) - 0.08w$, and $r^*(s, \text{exit}) = 0$, where $u \in \{0.5, 1.5, 3.0\}$. Analyses A through C and E through F are evaluated at the population level with no estimation noise. Analysis D uses finite-sample estimation with the actual IQ-Learn and GLADIUS implementations from the `econirl` package.

5.1 A. Reward recovery

Methods with anchors or Bellman consistency pin the value function offset and recover r^* exactly. The reduced-form logit converges to A^*, which bundles reward with continuation values. AIRL without anchors recovers a shaped perturbation with $\delta = 0.5V^*$. All five methods produce Type I CCP errors below 10^{-6}, confirming Proposition 6.

Method	MSE vs. r^*	Corr. with r^*
Oracle / AIRL+anchors / IQ-Learn / GLADIUS	0.000	1.000
AIRL-no-anchors	10.005	0.440
Reduced-form	40.020	0.138

Table 1: Population-level reward recovery.

5.2 B. Type II error vs. transition shift

The counterfactual changes the transition kernel so that buying advances k episodes instead of one. The Bellman equation is re-solved under each method's recovered reward. The reduced-form approach cannot separate reward from continuation values, so it re-solves with A^* as a proxy for reward, which embeds the original transition structure. AIRL without anchors re-solves with the shaped reward g^*, which embeds the shaping residual.

k	Oracle	AIRL+a	IQ/GLADIUS	AIRL-no-a	RF
1	0.000	0.000	0.000	0.000	0.000
2	0.000	0.000	0.000	0.025	0.082
3	0.000	0.000	0.000	0.017	0.094
5	0.000	0.000	0.000	0.015	0.123
7	0.000	0.000	0.000	0.016	0.129
10	0.000	0.000	0.000	0.014	0.139

Table 2: Type II CCP error vs. oracle. All five methods are shown. Structural methods with correct reward recovery are exact at every k.

5.3 C. Shaping magnitude sweep

We construct shaped rewards $\tilde{r} = r^* + \alpha V^*(s) - \alpha\beta V^*(s')$ for $\alpha \in [0, 1]$ and evaluate the Type II counterfactual with $k = 3$. At $\alpha = 0$ the reward is exact and at $\alpha = 1$ it is maximally shaped.

α	Type II CF error	Corr. with r^*
0.00	0.000	1.000
0.10	0.004	0.908
0.30	0.011	0.610
0.50	0.017	0.440
1.00	0.027	0.270

Table 3: Counterfactual error is monotone in shaping magnitude. Without anchors, which α the optimizer finds is indeterminate.

5.4 D. Finite-sample convergence

We compare five finite-sample estimators across sample sizes from 200 to 10,000 trajectories (3 seeds each). Reduced-form logit targets A^*, which is the wrong objective for Type II counterfactuals. IQ-Learn (χ^2) optimizes the Garg et al. (2021) tabular objective with L-BFGS-B. AIRL trains an adversarial discriminator without anchor constraints, so shaping ambiguity persists in finite samples as in Analysis A. LSW-AIRL enforces the two anchor constraints (exit action 2 pinned to zero reward, absorbing state s_{abs} pinned to zero value), achieving full identification at population. GLADIUS uses neural Q and EV networks with 150 training epochs.

N	RF	IQ-Learn (χ^2)	AIRL	LSW	GLADIUS
200	0.265	0.237	–	–	0.132
500	0.264	0.287	–	–	0.135
2,000	0.263	0.285	–	–	0.139
5,000	0.262	0.335	–	–	0.135
10,000	0.262	0.333	–	–	0.130

Table 4: Type II CF error ($k = 3$, 3 seeds). Results pending re-run with all five estimators.

5.5 E. Anchor misspecification

We set the true exit payoff to ε while the anchors assume it is zero. All five methods are evaluated at the population level. The reduced-form and AIRL-no-anchors errors are not affected by the anchor misspecification because they do not use anchors. The error shown is for AIRL with anchors, which degrades smoothly.

ε	Corr. with r^*	Type II CF error	In-sample π shift
0.0	1.000	0.000	0.000
0.5	0.976	0.002	0.000
1.0	0.909	0.004	0.000
2.0	0.729	0.008	0.000

Table 5: Anchor misspecification for AIRL with anchors. Degradation is smooth. In-sample CCPs are unaffected because the policy is identified regardless of reward levels.

5.6 F. Content-based paywall

High-content episodes become paid-only and the wait action is blocked, forcing buy-or-exit. This is a Type II shift that mirrors Section 7.3 of LSW. All five methods are evaluated at the population level.

Method	CCP error
Oracle	0.000
AIRL+anchors	0.000
IQ-Learn / GLADIUS	0.000
AIRL-no-anchors	0.009
Reduced-form	0.185

Table 6: Content-based paywall counterfactual. The reduced-form approach fails at 18.5pp because it cannot separate content utility from continuation value.

5.7 G. State-only reward (Fu et al. transfer test)

Analyses A through F use action-dependent rewards, which is the LSW setting. Here we test the Fu et al. (2018) setting by replacing the true reward with a state-only function $r^*(s, a) = r^*(s)$ for all a, where $r^*(s) = u(c)$ depends only on content quality. This satisfies the conditions of Fu et al. Theorem 5.1, so AIRL with a state-only $g_\theta(s)$ should recover $r(s)$ up to a constant that cancels in the softmax. We evaluate the Type II counterfactual with $k = 3$, re-solving the Bellman equation without pinning $V(s_{\text{abs}}) = 0$ (the absorbing-state anchor is not part of the Fu et al. identification strategy).

Method	MSE vs. r^*	Corr.	Type II err
Oracle / AIRL+anchors / IQ-Learn	0.000	1.000	0.000
AIRL-state-only ($g^* = r + \text{const}$)	2.778	1.000	0.000
AIRL-shaped ($\alpha = 0.5$)	13.689	0.530	0.026
Reduced-form	54.755	0.000	0.182

Table 7: State-only reward DGP. AIRL with state-only $g_\theta(s)$ recovers $r^*(s)$ up to a constant (MSE > 0 but Corr $= 1$) and transfers perfectly (Type II error $= 0$). This is the mechanism behind the Fu et al. sim-to-real experiments. Generic AIRL without the state-only restriction still fails (0.026) because the shaping residual is action-dependent. The reduced-form advantage has zero correlation with $r^*(s)$ because the true reward is identical across actions and the advantage reflects only continuation-value differences.

6 Summary

	RF	AIRL (no anchors)	AIRL (anchors)	IQ-Learn	GLADIUS
Uses $P(s'\|s,a)$	No	No (sim.)	No (sim.)	Yes (exact)	No (empirical)
Recovers	A^*	r + shaping	r^*	r^*	r^*
Type I CF	✓	✓	✓	✓	✓
Type II CF	×	×	✓	✓	✓

Table 8: Summary of five approaches to reward recovery.

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